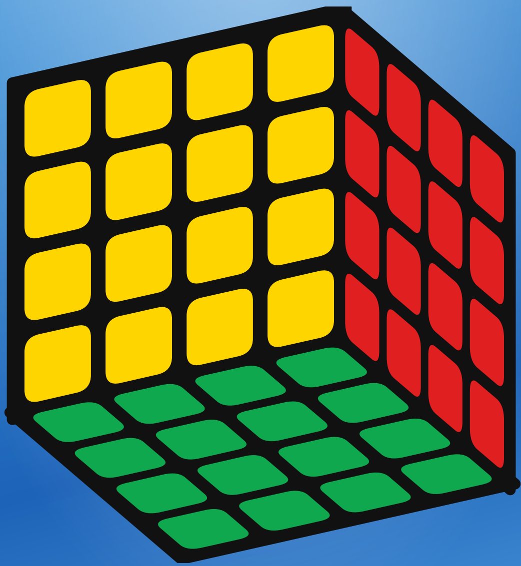
4×4 CUBE

6 SIDES · 24 CENTERS · 24 EDGES · 8 CORNERS

SOLUTION GUIDE



SOLVE IT LIKE A 3×3!

STAGE 1:

GET TO KNOW YOUR 4×4 CUBE

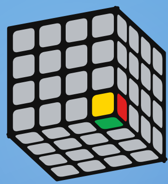
THE BIG IDEA

A 4×4 has no fixed centers and its edges are split into **two halves**. The trick is **reduction**: first build the center blocks and join the edge halves, and then the whole puzzle behaves **exactly like a 3×3** — so you finish it with the same moves you already know.

THE PARTS:

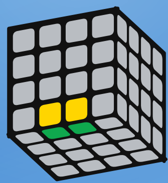
CORNER PIECES

Three (3) colors. There are **eight (8)**, one at each corner of the cube — just like a 3×3.



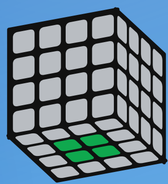
EDGE PIECES (WINGS)

Two (2) colors. There are **24**, in matching pairs. Two wings of the same two colors make one 3×3 edge once you pair them.



CENTER PIECES

One (1) color, four to a face (**24** total). They are **not** fixed — you build each face's 2×2 center block yourself.



WHITE and **YELLOW** are on opposite faces. We build the white side first and finish on the yellow side. (Use YOUR cube's colors; the steps are the same.)

EACH FACE TURN HAS A LETTER

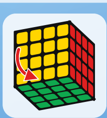
A letter alone (like **(R)**) means turn that face a quarter turn **clockwise** as you look straight at it. A letter with an apostrophe (**(R)**') means **counter-clockwise**. A letter with a **2** (R2) means a half turn. **U**p, **D**own, **R**ight, **L**eft, **F**ront, **B**ack.

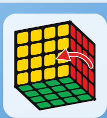
For the very last "parity" page you will also use **wide** turns, written like **Rw** — turn the face **and** the slice next to it together (two layers at once).

VERY IMPORTANT

Hold your cube to match each picture before you start a sequence. Dark gray in a picture means **that color does not matter**.


U


U'

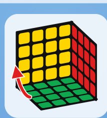

D


D'


R


R'


L


L'


F


F'


B


B'

STAGE 2: BUILD THE 6 CENTERS

HOLDING YOUR 4×4: Pick a color to start (we use white). Gather its four center pieces into one 2×2 block on a face. Then do the same for every color — opposite colors go on opposite faces.

TIPS:

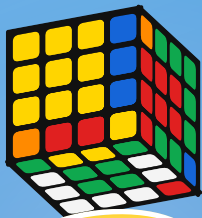
- Use the inner slice turns (the middle layers) to bring matching center pieces onto the same face, then park them together.
- Finish a face, then hold it on the side or bottom so later slice turns do not break it.
- When all six centers are solid blocks, the centers are done — their colors now tell you where every other piece belongs.

STAGE 3: PAIR THE EDGES

HOLDING YOUR 4×4: Find two edge wings that share the same two colors and join them so they sit side by side as one edge. Repeat until all 12 edges are paired.

TIPS:

- Bring one wing to the top-front and its matching wing next to it, then turn a wide layer to lock them together — and turn it back so the centers stay solid.
- Keep finished pairs out of the way (in the top or bottom) while you join the next pair.
- Once all 12 edges are joined, STOP using slice and wide turns. From here, only turn whole faces — your 4×4 is now a 3×3!



GOAL: 12 JOINED EDGES

STAGE 4: MAKE THE WHITE CROSS

HOLDING YOUR 4×4: Turn the cube so **WHITE** is on the bottom. Make a white plus on the bottom, and make each edge's side color match its center.

TIPS:

- Find a white edge, line it up under the matching side center, then turn that face twice to drop it into the cross.
- If a white edge is flipped, take it to the top first, then bring it down the right way.
- This is exactly the 3×3 white cross — only whole-face turns from now on.

STAGE 5: FINISH THE WHITE LAYER

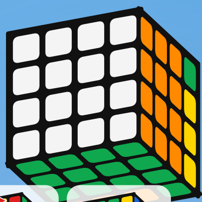
HOLDING YOUR 4×4: Put the four white corners in place to complete the bottom layer, white on the bottom and side colors matching.

TIPS:

- Find a white corner in the top layer, hold it above the empty corner where it belongs.
- Use **(R)** **(U)** **(R)'** (and a top turn between tries) to drop it in with white on the bottom.

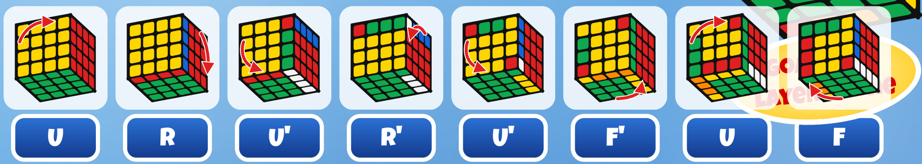
STAGE 6: SOLVE THE MIDDLE EDGES

HOLDING YOUR 4×4: Flip the cube so white is on the bottom and yellow on top. Place the four middle-layer edges (the ones with no yellow).



TIPS:

- Find a top edge with no yellow. Turn the top so its front color matches a side center, making a sideways T.
- If it belongs to the right, do (U) (R) (U)' (R)' (U)' (F)' (U) (F). Mirror it for the left.

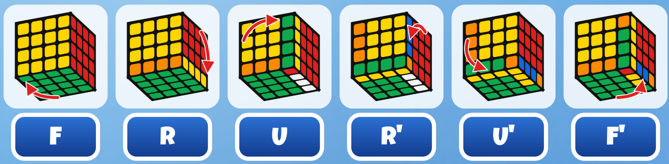


STAGE 7: YELLOW CROSS & TOP

HOLDING YOUR 4×4: With yellow on top, make a yellow cross, then make the whole top face yellow.

TIPS:

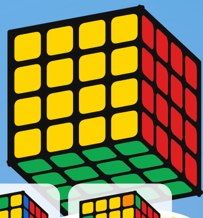
- For the cross, repeat (F) (R) (U) (R)' (U)' (F)' until the yellow plus appears.
- Then orient the yellow corners (turn the top so an unsolved corner is front-right and repeat a corner move) until the whole top is yellow.



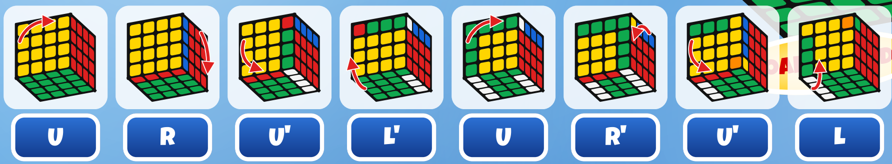
STAGE 8: FINISH THE LAST LAYER

HOLDING YOUR 4×4: Move the last pieces into place: first the corners, then the edges, until the cube is solved.

TIPS:



- Position the yellow corners using a corner 3-cycle, turning the top between tries so the solved corners stay put.
- Then cycle the last edges into place. These are the same last-layer moves as the 3×3.



STAGE 9: PARITY (4×4 ONLY)

Because a 4×4 has split edges, you can reach two cases a 3×3 never shows. They are not mistakes — just do the matching fix and keep going.

ONE FLIPPED EDGE (OLL)

If a single last-layer edge looks flipped while making the yellow cross, do:

Rw2 B2 U2 Lw U2 Rw' U2 Rw U2 F2 Rw F2 Lw' B2 Rw2

TWO SWAPPED EDGES (PLL)

If only two last-layer edges need to swap at the very end, do:

2R2 U2 2R2 Uw2 2R2 Uw2

2R2

U2

U2

2R2

UW2

2R2

UW2

CONGRATULATIONS!

You reduced a 4×4 to a 3×3 and solved it! Centers, then edges, then the same moves you know. Now mix it up and do it again.

Every picture in this booklet was drawn by a computer model of the cube that checks every move really works.